

Automorphisms of $\mathbb{Q}_p$

We prove that a field automorphism of $\mathbb{Q}_p$ must be identity.

Suppose $\tau : \mathbb{Q}_p \rightarrow \mathbb{Q}_p$ is a field automorphism.

Since $\tau(1) = 1$ and τ preserves addition and multiplication, it follows easily that $\tau(r) = r$ for every rational number r .

Given any p -adic number $x \in \mathbb{Q}_p$, there is a sequence (r_n) of rational numbers converging to it. This means that for every positive integer k , there exists an N_k such that for all $n \geq N_k$ we have $(r_n - x) \in p^k \mathbb{Z}_p$.

Suppose we prove that $\tau(p^k \mathbb{Z}_p) = p^k \mathbb{Z}_p$ for all k .

Then $r_n - \tau(x) = \tau(r_n - x)$ would also be in $p^k \mathbb{Z}_p$. This would prove that (r_n) converges to $\tau(x)$; which would prove that $\tau(x) = x$.

Divisibility of $1 + p^k \mathbb{Z}_p$

As seen earlier, we have a group isomorphism $\exp : p^k \mathbb{Z}_p \rightarrow 1 + p^k \mathbb{Z}_p$ for $k \geq 1$ for all $p > 2$, and for $k \geq 2$ for $p = 2$. (Recall that $\exp(x) = 1 + \sum_{k=1}^{\infty} x^k / (k!)$ converges in these cases.) Here we treat $p^k \mathbb{Z}_p$ as an additive group and $1 + p^k \mathbb{Z}_p$ as a multiplicative group.

Given an integer a such that $\gcd(a, p) = 1$, the rational number $1/a$ lies in $\mathbb{Z}_p$. This shows that multiplication by a is an invertible homomorphism from $\mathbb{Z}_p$ to itself. Consequently, the same is true for the isomorphic group $p^k \mathbb{Z}_p$. We use the above isomorphism to conclude:

The map $x \mapsto x^a$ is a group automorphism of the group $1 + p^k \mathbb{Z}_p$ if $\gcd(a, p) = 1$.

For a positive integer a and an Abelian group G , when the map $x \mapsto x^a$ from G to itself is surjective, we say that G is *a-divisible*. When it is an isomorphism, we say that G is *uniquely a-divisible*. Thus, we can re-state the above:

$1 + p^k \mathbb{Z}_p$ is uniquely a -divisible when a satisfies $\gcd(a, p) = 1$.

Divisible subgroups of $\mathbb{Q}_p^*$

As seen earlier $\eta : (n, a, x) \mapsto p^n \omega(a) \exp(x)$ gives an isomorphism from $\mathbb{Z} \times (\mathbb{Z}/\langle p^k \rangle)^* \times p^k \mathbb{Z}$ to $\mathbb{Q}_p^*$ when:

- $p > 2$ and $k = 1$
- $p = 2$ and $k = 2$

Given a property P of groups, and a group automorphism of a group G , it is clear that if there is a largest subgroup of G that has the property P , then it is preserved by the automorphism.

When $p > 2$ the largest $(p-1)$ -divisible subgroup of $\mathbb{Z} \times (\mathbb{Z}/\langle p \rangle)^* \times p\mathbb{Z}$ is $\{0\} \times \{1\} \times p\mathbb{Z}_p$. Using the isomorphism given by η , we see that any group automorphism of $\mathbb{Q}_p^*$ preserves the subgroup $1 + p\mathbb{Z} = \eta(p\mathbb{Z}_p)$.

In particular, we see that $\tau(1 + p\mathbb{Z}_p) = 1 + p\mathbb{Z}_p$. Since $\tau(1) = 1$, it follows that $\tau(p\mathbb{Z}_p) = p\mathbb{Z}_p$ for $p > 2$.

When $p = 2$ multiplication by $3 = (2^2 - 1)$ gives an automorphism of the subgroup $\{0\} \times \{\pm 1\} \times 4\mathbb{Z}_2$ and this is the largest subgroup of $\mathbb{Z} \times \{\pm 1\} \times 4\mathbb{Z}_2$ on which this is true. This means that τ preserves the image under η of this subgroup in $\mathbb{Q}_2^*$. One easily checks that this image is $1 + 2\mathbb{Z}_2$.

Thus, $\tau(1 + 2\mathbb{Z}_2) = 1 + 2\mathbb{Z}_2$. Since $\tau(1) = 1$, it follows that $\tau(2\mathbb{Z}_2) = 2\mathbb{Z}_2$.

In both cases, we see that τ preserves $p\mathbb{Z}_p$.

Conclusion

Since τ preserves $p\mathbb{Z}_p$ and $\tau(p) = p$, we see that τ preserves $p^k\mathbb{Z}_p$ for every integer k .

As seen above, this means that τ takes sequences of rational numbers that converge in $\mathbb{Q}_p$ to convergent sequences. Since τ is *identity* on rational numbers, we see that τ is the identity map on $\mathbb{Q}_p$ as required.